Course Introduction

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Having a challenging time to pronounce my name?

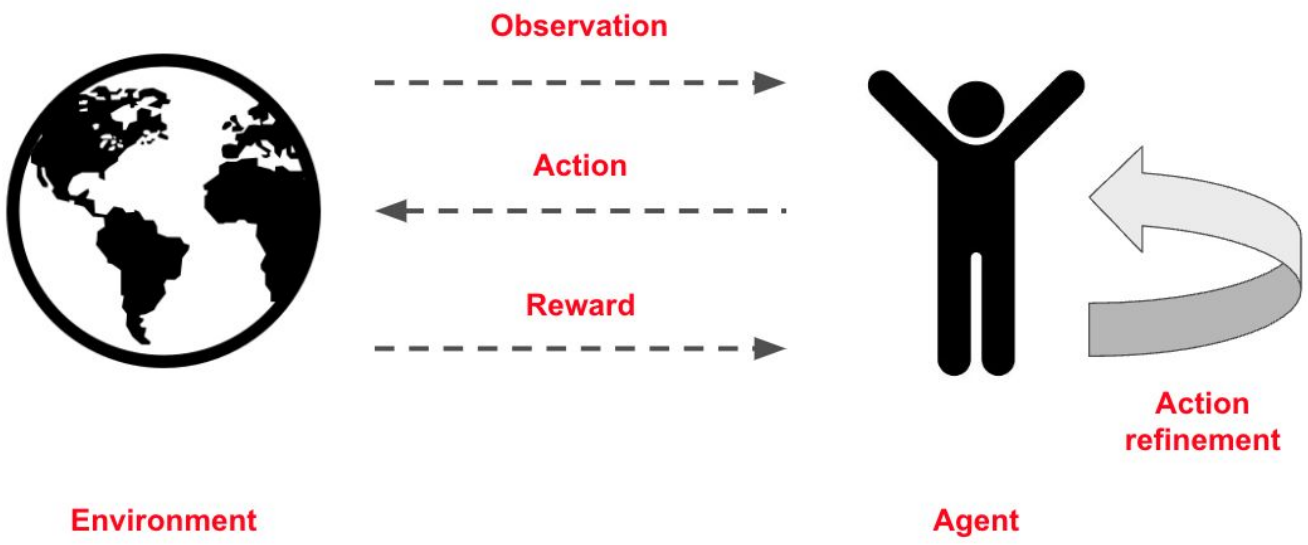
San José State University

CS272.01 Reinforcement Learning and Sequential Decision Making

Introduction to reinforcement learning, deep reinforcement learning, other online learning algorithms, and their applications.

This course is…

- CS 272 Reinforcement Learning and Sequential Decision Making
- RL is an area of artificial intelligence
- An agent learns a sequence of optimal actions to realize a goal

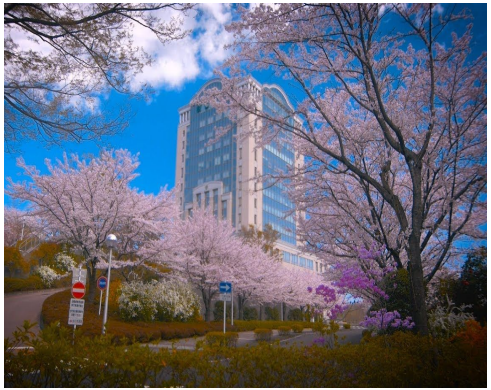


Today’s Agenda

- My Introduction
  - Breakout: Getting to know new fellow students
- My research areas
- Course syllabus
- Announcements
- Q&A

About Me

- Born and raised in Japan
- Obsessed with LEGO during the childhood
- Majoring CS from my bachelors
  - M.S. & B.S. in Engineering from Soka University
- Came to US for my PhD in 2016
  - Ph.D. & M.S. in CS from University of Texas at Dallas
- Joined SJSU in August, 2021



Breakout: Getting to know new fellow students

- Form a group (with 4 or 5 ppl around you)
- Introduce yourself!

About Me: Research Interest

- 5G and Beyond Networking & Network Slicing
- Deep Reinforcement Learning

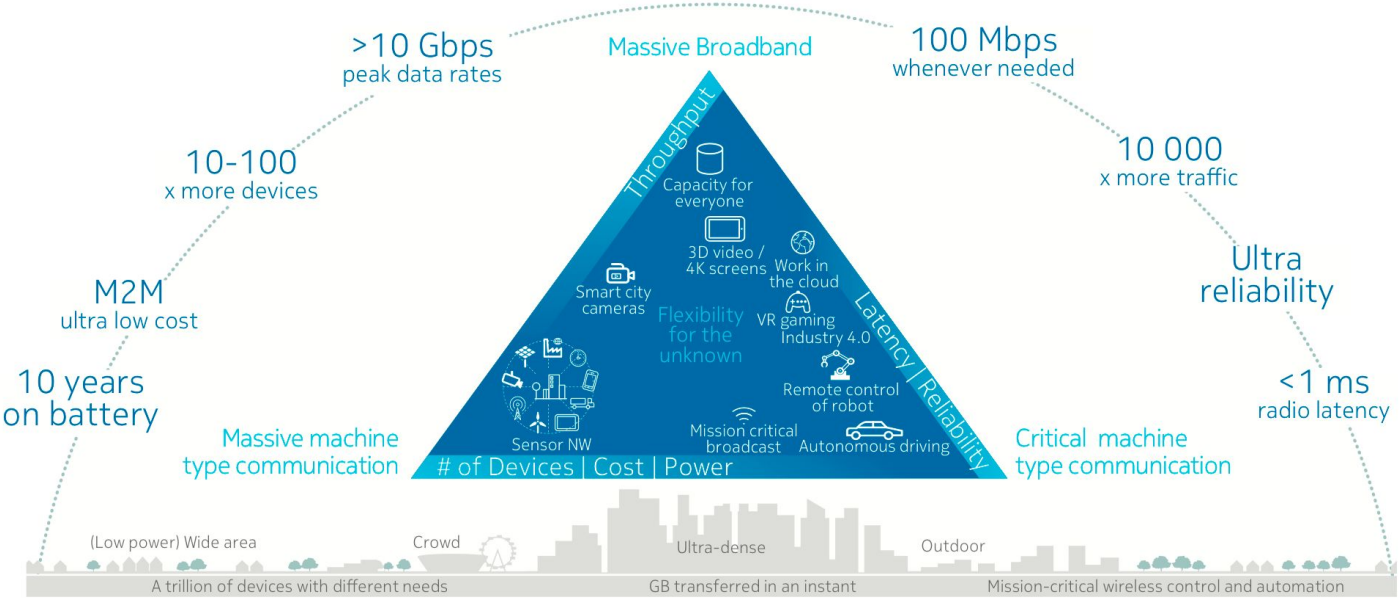


Figure 1. 5G will enable very diverse use cases with extreme range of requirements

About Me: Research Interest

- 5G and Beyond Networking & Network Slicing
- Deep Reinforcement Learning

Documentary: “AlphaGo - The Movie”, https://www.youtube.com/watch?v=WXuK6gekU1Y

For more information

Google

sjsu interconnect

All | News | Images | Maps | Shopping | More | Tools

About 119,000 results (0.52 seconds)

https://sjsu-interconnect.github.io

interconnect lab | SJSU interconnect lab researches next ...

SJSU interconnect lab researches next-generation telecommunication networks, with an emphasis on developing combinatorial optimization and machine learning ...

2024 Turing Award!

- ACM named **Andrew Barto and Richard Sutton** as the recipients of **the 2024 ACM A.M. Turing Award** for developing the conceptual and algorithmic foundations of reinforcement learning
- In a series of papers beginning in the 1980s, Barto and Sutton introduced the main ideas, constructed the mathematical foundations, and developed important algorithms for reinforcement learning—one of the most important approaches for creating intelligent systems.

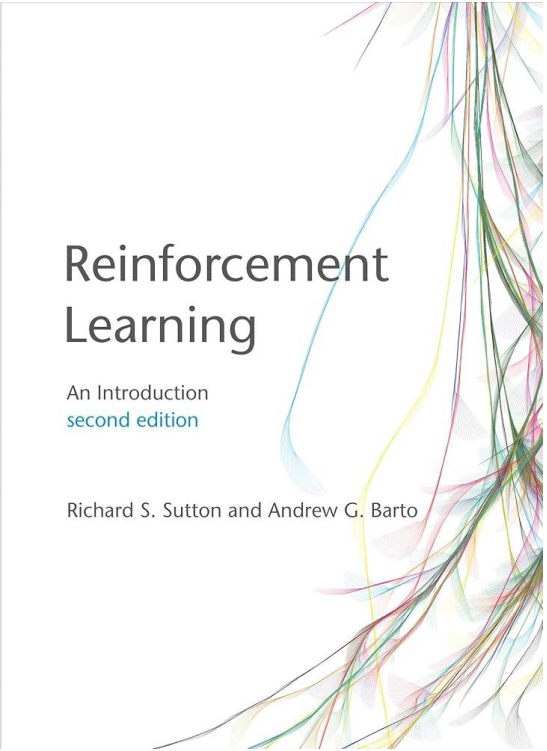
Andrew Barto

Dr. Richard Sutton

https://amturing.acm.org/

Hear from them: https://www.youtube.com/watch?v=PI0QJr3se_U

Our Textbook



Andrew Barto

Dr. Richard Sutton

http://incompleteideas.net/book/the-book-2nd.html

Course Syllabus

Check Canvas!

Wednesday Quizzes

- 8 quizzes (Please check the syllabus for the schedule)
  - The worst three will be dropped
  - No retake or makeup
- Canvas Online Exam (**Lockdown browser required**)
- Basic questions about the concepts and coding
- A review session right after the quiz during the class
  - Questions/Answers will NOT be published on Canvas
  - Please take notes during the review
- An opportunity to check your understanding and ask questions

Programming Assignment Format

- We will do coding together in class
  - Ready to bring your laptop with
    - A Python environment (See the next slide)
    - Zoom to share my live coding
  - Laptop lending at Student Computing Services
- You need to complete the rest after the class by yourself
- A submission may be evaluated based on python unit testing
  - Actual test cases will not be revealed
  - A submission needs to follow a specified format for the auto evaluation
    - If your code cannot be run due to the format issue, the submission may not be correctly evaluated
    - Also, the quality of the codes will be evaluated

Python Reference

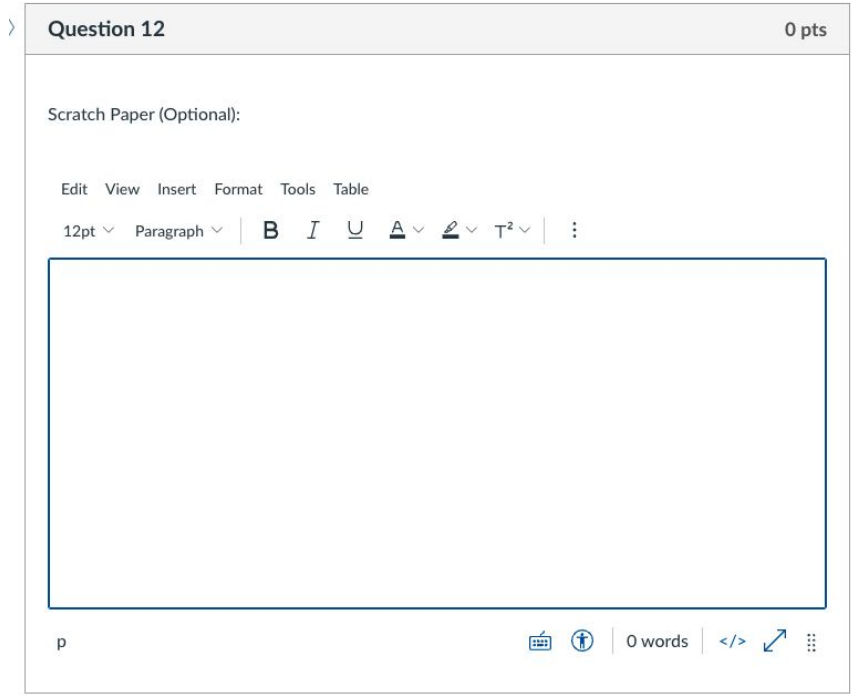
- OOP in Python: https://realpython.com/python3-object-oriented-programming/
- Virtual python environment (make sure you have Python 3.x)
  - venv: https://docs.python.org/3/library/venv.html
  - Anaconda: https://www.anaconda.com/
  - mini-forge (especially for M1 Mac): https://github.com/conda-forge/miniforge
- IDE
  - PyCharm: https://www.jetbrains.com/pycharm/
  - VSCode: https://code.visualstudio.com/docs/languages/python
- My environment: VSCode + Python 3.10.5 (through mini-forge)

Programming Assignment Warnings

- Code similarity will be checked using a plagiarism software
  - If the score exceeds 20%, you will receive an F grade for the assignment
- You can discuss the problem with other fellow students
- Never show/share your codes
- Do not write codes (or maybe pseudo codes) with fellow students

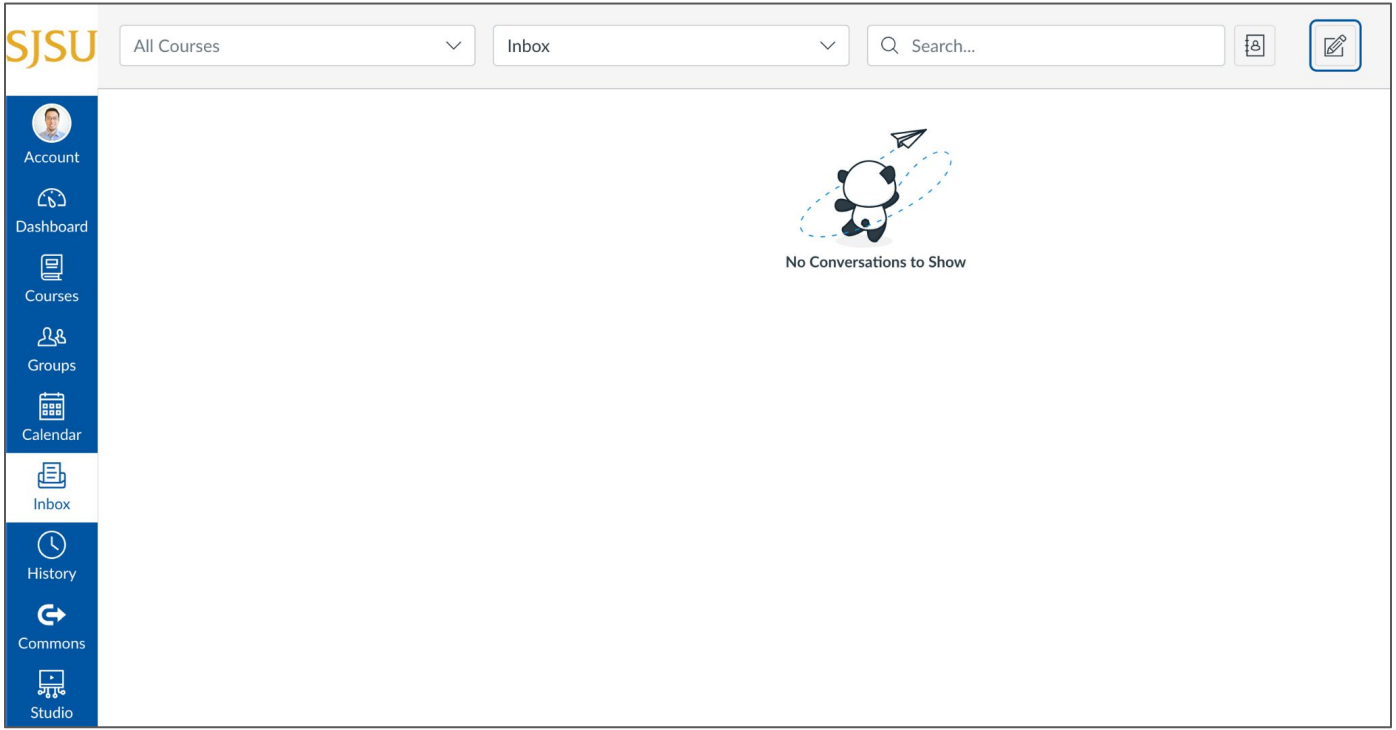
Exam Format

- Canvas Online Exam in Class (Lockdown browser required)
- Conceptual questions (similar to those in Wednesday Quizzes)
- Calculation questions (review in-class exercises)
- Scratch paper section for partial credits
- No cumulative final



Communication

Please use Canvas Message



Communication

No Email!

Any Questions?